Soc Internship day 27

1. Explain EDR and XDR

EDR (Endpoint Detection and Response)

EDR is a cybersecurity solution that continuously monitors endpoint devices such as laptops, desktops, servers, and mobile devices to detect, investigate, and respond to threats.

Functions:

- Real-time endpoint monitoring
- Threat detection and alerting
- Incident investigation
- Malware detection and containment
- Automated response actions

Example: If ransomware starts encrypting files on a laptop, EDR can detect the suspicious activity and isolate the device from the network.

XDR (Extended Detection and Response)

XDR extends security visibility beyond endpoints by collecting and correlating data from multiple security layers such as:

- Endpoints
- Network traffic
- Email systems
- Cloud environments
- Identity systems

Functions:

- Centralized threat detection
- Cross-platform correlation
- Automated investigation
- Faster incident response
- Improved visibility across the organization

Example: A phishing email compromises a user account, and the attacker later accesses cloud resources. XDR can connect these events and identify the complete attack chain.

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2. EDR vs XDR

Feature	EDR	XDR
Scope	Endpoints only	Multiple security layers
Visibility	Device-focused	Organization-wide

Feature	EDR	XDR
Data Sources	Endpoint logs	Endpoint, network, email, cloud, identity
Threat Correlation	Limited	Advanced cross-platform correlation
Investigation	Endpoint-centric	Full attack chain visibility
Response	Endpoint actions	Coordinated responses across systems
Complexity	Simpler	More comprehensive

Summary:

- **EDR** protects and monitors endpoints.
- **XDR** provides broader security visibility and correlates events across the entire environment.

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3. Malware Detection Scenario

Scenario

An employee receives a phishing email containing a malicious attachment.

Attack Steps

1. User opens the attachment.
2. Malware executes on the endpoint.
3. Malware downloads additional payloads from a malicious server.
4. Malware attempts to steal credentials and spread laterally.

EDR Detection

- Detects unusual process execution.
- Identifies malicious file behavior.
- Alerts security team.
- Isolates the infected endpoint.

XDR Detection

- Detects the phishing email.
- Correlates email event with endpoint infection.
- Identifies suspicious network connections.
- Detects credential abuse attempts.
- Provides a complete attack timeline.

Response

1. Isolate affected device.
2. Block malicious IPs/domains.
3. Remove malware.
4. Reset compromised credentials.
5. Review logs for lateral movement.
6. Update security controls and user awareness training.

Result: EDR stops the malware on the device, while XDR provides visibility into the entire attack lifecycle and helps prevent further compromise.